

Filling process

Filling in 120 bpm (bottles per minute):

- 120 bpm
- 20 – 20 – 6
- 140 bpm
- 24 – 24 – 8
- Filling
- Rinser filler capper (line) 2016 model (??)

Other parameters:

- Closure torque – 3.8 Nm
- CCP-2 – control limits – 84°C – 88°C
- Target – 86°C
- OPRP-4 — 10.0 – 18.0 lbs (underlined)
- Target – 14.0 lbs
- Hot filling – 83°C
- Warm filling – 71°C
- 400 ml volume bottle
- OPRP-4 – 10 – 18 lb-ft (underlined)
- Operational prerequisite: Capping torque
- Ensure safety – between PRP & CCPs

Filling flow chart

Preforms

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Blowing

↓

Heating

↓

Rinsing

↓

Filling

↓

Neck Rinsing – Bottle threads

↓

Capping – Closure

- PRP – prerequisite program
- OPRP – operational prerequisite program
- CCP – critical control point

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- **Return Juice Temperature** → **80°C**
 - **Buffer Tank Level** → (maintained at set level for continuous flow)

300 BPM (Bottles Per Minute)

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- closure /sapper – 28 mm

CCP

- CCP-2 – control limits – 84°C – 88°C
- Target – 86°C (86 ± 2) (“86±2” is underlined)
- OPRP-4 – (capping torque) – 10.0 – 18.0 lbs
- Target – 14 lbs (written as “Target – 14 lbs”)

Filling temperatures / bottle:

- Hot filling – 83°C
- Warm filling – 71°C
- 400 ml volume bottle
- OPRP-4 – 10–18 lb-ft (lb-ft is underlined)

- operational prerequisite program – Capping torque
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Filling flow chart

Preforms

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Blowing

↓

Heating

↓

Rinsing

↓

Filling

↓

Neck Rinsing – Bottle threads

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Capping – Closure

CIP – Clean in place

- Caustic soda – 95°C ; Return – 85°C
- Orthophosphoric acid

7 steps:

1. Pre-rinsing – 15 min
 2. Caustic
 3. Caustic rinsing
 4. Orthophosphoric
 5. Orthophosphoric circulation
 6. Hot water
 7. Final rinsing
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COP – cleaning on place

Food safety interlocks

- Bottle rinsing pressure – 2 bar
 - Hot water temp – 80°C
 - Filling temp – 86 ± 2
 - Return temp – 80 ± 2
 - Neck rinsing water – 80
 - Cap rinsing – 0.5 to 1 bar
 - Tank pressure – 0.1
 - Rinsing water – 80°C
 - Filling temp – 86 ± 2
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- Return juice temp – 80
- Buffer tank level

300 BPM

- Ionized pressure cleaning
- Rinsing is not applicable
- UV used in hopper
- Capper line unit
- 28 mm – cap size
- Cap torque – 10–18 lbs (unit written as “lbs”)

Filling / measurement:

- Volumetric filling
 - Flow meter – pulse: 1 pulse = 2 ml
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CIP

5-step CIP:

1. Pre-rinsing – 10 min
2. Caustic circulation – 15 min

3. Caustic RO water rinsing – 10 min (noted 85°C)

4. Hot water circulation – 15 min

5. Final rinsing – 10 min

300 BPM – filling

- Hot filling
 - Temp – 85°C – 95°C
 - Temp controller is pasteurizer
 - Technology combine (combi)
 - PET
 - Each filling have 60 valves
 - 250 – 248 ml
 - Filler tank temp – 88.6°C
 - Neck rinsing – 80°C
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- 60 valves
- 20 – capping head
- U.V. rays
- Preheat temp – 85–95°C
- Rinsing temp – 80°C
- Pressure – 0.8
- Volumetric filling
- Automatic CIP process

140 bpm

- ... Rinser / Filler / Capper – 38 mm
- Filling temp – $86 \pm 2^\circ\text{C}$
- Return juice temp – $> 83^\circ\text{C}$

- Hot water temp – 80°C
- Bottle rinsing pressure – 2.0 bar (1.5–3 bar)
- Neck rinsing pressure – 2.0 bar (1.5–3 bar)
- Cap rinsing pressure – 0.5 bar
- UV light on

Hot filling / warm filling

- Hot filling – 86 ± 2
- Warm filling – 74 ± 2

CIP

- Hot water – 95°C
- Caustic solution – 15 min
- (Temp) – 83°C (written as “83°C” on the next line)
- Rinser – 15–20 (unit not specified; likely seconds/minutes, but not written clearly)
- Neck rinsing box
- Pressure – 2–6 bar
- Cap with UV treat (written as “Cap with UV treat”)

“Valves” list :

Valves:

- V1 – 2-way piston line valve
- V2 – filler tank feed line valve
- V3 – filler tank spray line valve
- V4 – buffer tank spray (sprinkler)
- V5 – filler tank exhaust line valve
- V6 – filler tank drain line valve
- V7 – buffer tank drain line valve1
 - V8 – filler tank air inlet value
 - V9 – 3-way return line value